

Theory of computation

① Automata theory

// what DFAs / NFAs / PDAs or other special computers can or cannot do.

② computability theory

// What a general computer can or cannot do.

③ Computational complexity

// what a computer can or cannot do efficiently.

The diagonalization method

(G. cantor 1873)

— How do we measure size of a set?

- i) finite set : Just count
- ii) Infinite set : We can define how to count an infinite set.

We will first go through some definitions.

Def

$f: A \rightarrow B$

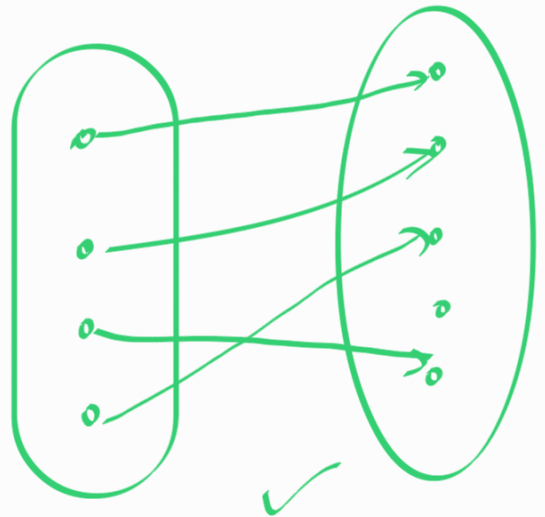
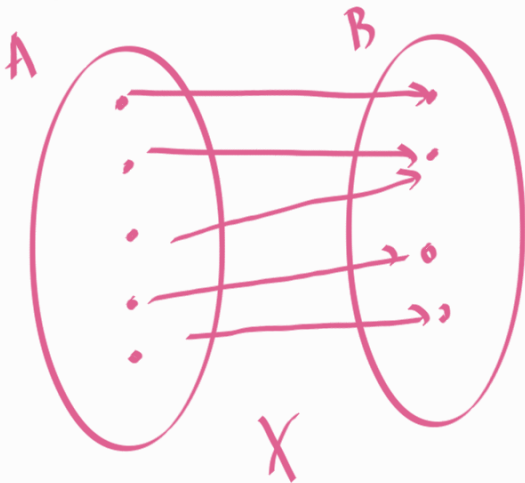
Domain $\swarrow$ A $\searrow$ Co-domain B

- (i) $\forall a \in A: f(a)$ is defined
- (ii) $\forall a \in A: f(a)$ produces an unique value
- (iii) $\forall a \in A: f(a) \in B$

Def

Given function $f: A \rightarrow B$,
we say that f is one-to-one
(1:1, injective) if,

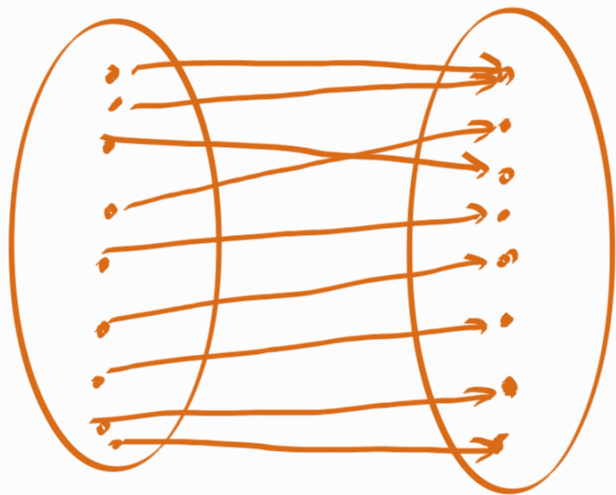
- (i) $\forall a, b \in A: f(a) = f(b) \Rightarrow a = b$



Def

Given function $f: A \rightarrow B$,
we say that f is onto (surjective),
it,

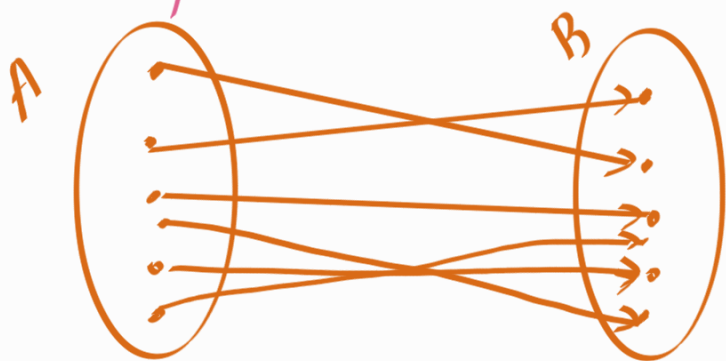
$$(11) \forall b \in B : \exists a \in A : f(a) = b$$



Def

Given function $f: A \rightarrow B$,
we say that f is a correspondence
(bijective), if f is onto and one-to-one.

Note that if $f: A \rightarrow B$ is a
correspondence, then $|A| = |B|$



Def

A set A is countable if it is finite or it has the same size as $\mathbb{N} = \{1, 2, 3, 4, \dots\}$.

Theorem

Every subset of a countable set is countable.

Proof is omitted.

Let's look at an example.

$$E = \{e \in \mathbb{N} : e \% 2 = 0\}$$

I want to show that E is countable.
we need to show that $\exists$ a bijection

$$f: \mathbb{N} \rightarrow E$$

$$f(n) = 2n$$

$$\begin{array}{ccccccccccc} E = \{ & 2 & , & 4 & , & 6 & , & 8 & , & 10 & , & 12 & , & \dots & \} \\ & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \\ \mathbb{N} = \{ & 1 & , & 2 & , & 3 & , & 4 & , & 5 & , & 6 & , & \dots & \} \end{array}$$

Ex 2: $Q^+ = \{ m/n : m, n \in \mathbb{N} \}$

- we can show a correspondence between Q^+ and $\mathbb{N}$
- one way to do this is to list all elements of Q^+ .
- we should list all elements of Q^+ exactly once
- Then we can pair up 1st element with 1, 2nd element with 2, and so on.

$$m+n=k$$

	1	2	3	4	5	6	7	...
1	$\frac{1}{1}$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{6}$	$\frac{1}{7}$	
2	$\frac{2}{1}$	$\frac{2}{2}$	$\frac{2}{3}$	$\frac{2}{4}$	$\frac{2}{5}$	$\frac{2}{6}$	$\frac{2}{7}$	
3	$\frac{3}{1}$	$\frac{3}{2}$	$\frac{3}{3}$	$\frac{3}{4}$	$\frac{3}{5}$	$\frac{3}{6}$	$\frac{3}{7}$	
4	$\frac{4}{1}$	$\frac{4}{2}$	$\frac{4}{3}$	$\frac{4}{4}$	$\frac{4}{5}$	$\frac{4}{6}$	$\frac{4}{7}$	
5	$\frac{5}{1}$	$\frac{5}{2}$	$\frac{5}{3}$	$\frac{5}{4}$	$\frac{5}{5}$	$\frac{5}{6}$	$\frac{5}{7}$	
6	$\frac{6}{1}$	$\frac{6}{2}$	$\frac{6}{3}$	$\frac{6}{4}$	$\frac{6}{5}$	$\frac{6}{6}$	$\frac{6}{7}$	

$\{ \frac{1}{1}, \frac{1}{2}, \frac{2}{1}, \frac{1}{3}, \frac{3}{1}, \frac{1}{4}, \frac{2}{3}, \frac{3}{2}, \frac{4}{1}, \frac{1}{5}, \dots \}$

$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 $\{ 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, \dots \}$

$\therefore \mathbb{Q}^+$ is countable.

Ex: $\mathbb{R}$ is uncountable.

- In order to show that $\mathbb{R}$ is uncountable, we need to show that there exists no bijections between $\mathbb{R}$ and $\mathbb{N}$
- proof is by contradiction.

Assume that $\mathbb{R}$ is countable.

Then there exists a bijection

$$f: \mathbb{N} \longrightarrow \mathbb{R}$$

Then, we must be able to map each element in $\mathbb{N}$ to a unique element in $\mathbb{R}$ (injective), and every element in $\mathbb{R}$ must be mapped by this function f (surjective).

As an example f could be something like following.

n	$f(n)$
→ 1	2 <u>3</u> 1 4 6 8 9
→ 2	8 . 1 <u>1</u> 2 3 5 6 8 9 . . .
→ 3	0 . 0 0 <u>1</u> 2 3 1 2 3 . . .
→ 4	0 . 1 1 1 <u>2</u> 5 9 8 6 0
5	
⋮	
→ <u>i</u>	
⋮	

i

$$x = 2.1568 - - -$$

we construct $x \in \mathbb{R}$ such that each i^{th} decimal point in x differs from the real number mapped by bijection f .

Since, I can construct $x \in \mathbb{R}$ and we see that this value is not mapped by the bijection f .

$\therefore f$ cannot be a surjection
 $\therefore f$ cannot be a bijection.

$\therefore$ Initial assumption must be 'incorrect'.

$\therefore \mathbb{R}$ is uncountable.